# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 1 of 7 · Question 4 part c.i · 2 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a) | **The only correct answer is C**<br><br>***A*** *is not correct because* ***mebibits*** *is a measure of storage capacity*<br><br>***B*** *is not correct because* ***mebibytes*** *is a measure of storage capacity*<br><br>***D*** *is not correct because transmission speeds are measured only in bits per second, not* ***bytes per second*** | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(b) | Award **one** mark for each correct cell:<br><br>**Type** / **Example** / **Number of bits**<br>IPv4 / 192.169.0.3 / 32 (1)<br>IPv6 / 1050:a500:00c0:0440:0006:0300:700d:436f / 128 (1) | | 2 |

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 1 of 7 · Question 4 part c.i · 2 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

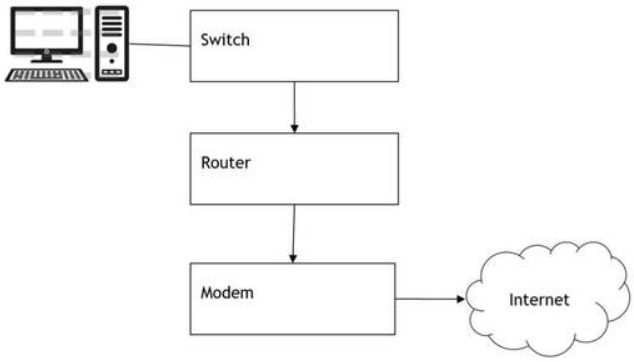
| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(i) | Award up to **two** marks for a linked description, such as:<br>• In a wired network packets travel over (copper/fibre-optic) cables (1), whereas in a wireless network packets travel over the air (1) | Allow radio signals, infra-red, microwave for air (must be more than just waves)<br><br>Allow twisted pair, CAT for copper.<br>Allow electrical/light signals (not just electricity) | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>1. A protocol (suite/group/family) (1)<br>2. Used on a wired network / wired connection (1)<br>3. Defines physical parts/type of cable(twisted pair, CAT6)/type of connector (1)<br>4. Defines how packets are checked for errors (1)<br>5. Defines the speed of transmission (1)<br>6. Operates at the link layer of the TCP/IP stack (1) | | 2 |

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 1 of 7 · Question 4 part c.i · 2 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(d) | Award **three** marks for all of switch, router and modem in the correct order as in diagram<br><br>Award **two** marks for all of switch, router and modem in any order<br><br>Award **one** mark for any of switch or router or modem in any box<br><br>  | | 3 |

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 1 of 7 · Question 4 part c.i · 2 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(e) | Award up to **two** marks for a linked description, such as:<br><br>Validation is used to check that the data/password entered by a user is fit for processing/meets specified requirements (1), whereas authentication is used to verify the identity of the user/prove that users are who they say they are (1) | For validation 'specified requirements' a good description of the check can be given e.g. presence check, length check, format check | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 2 of 7 · Question 3 part b · 1 mark

2019 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a)(i) | • LAN / Local area network (1)<br><br>Allow PAN/Personal Area Network | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(a)(ii) | • WAN / Wide area network (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(b) | A: Megabits per second<br><br>**The only correct answer is A**<br><br>**B** is not correct because megapixels are associated with images rather than network data speeds<br><br>**C** is not correct because mebicycles are not associated with network data speeds<br><br>**D** is not correct because mebihertz are not associated with network data speeds | | 1 |

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 2 of 7 · Question 3 part b · 1 mark

2019 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(c) | • IMAP (1)<br><br>AND **one** from:<br>• because changes are synced (1) with the mail server in real-time (1)<br>• because the messages always reside on the email server (1) as she has limited storage (1) on her tablet and phone<br>• it gets sent to both smartphone and tablet (1)<br><br>**Alternative answer:**<br>• POP3 (not awarded a mark)<br><br>AND **one** from:<br>• because storage limitations on server (1) and privacy issues (1)<br>• because more CPU time (1) may be used when syncing (1) (a large number of messages). | | 3 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(d) | | | 4 |

| Characteristic | Bus | Ring | Star | Mesh |
|---|---|---|---|---|
| Network performance degrades as more devices are added. | ✓ | | | |
| All devices are directly or indirectly connected to all others. | | | | ✓ |
| Each node has a physical attachment to a routing device. | | | ✓ | |
| There are no data collisions, because packets travel in the same direction. | | ✓ | | |

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 2 of 7 · Question 3 part b · 1 mark

2019 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(i) | -81<br><br>One mark for negative sign<br>One mark for 81. | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(ii) | One mark for each nibble in correct order<br><br>0100 1011 | | 2 |

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 2 of 7 · Question 3 part b · 1 mark

2019 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(iii) | Any **two** from:<br><br>1. This is an overflow (1) error<br>2. It is caused by adding two 1s in the most significant bit (1)<br>3. The resulting number is too large to fit in the number of bits available / requires 9 bits / the number is larger than $2^8$ / there is 1 that needs to be carried over (in MSB) / MSB gets discarded (1)<br>4. It can cause the program to crash or produce unreliable/incorrect results (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(a) | Any **two** from:<br>• (Share) access to the Internet/WWW/broadband connection (1)<br>• (Internal) communication (using email/instant messaging/calendars) (1)<br>• Share files/data (1)<br>• Share peripherals/printers / hardware (1)<br>• Increases the amount of storage (since network storage devices can be attached) (1)<br>• Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1)<br>• Centralised backup (1)<br>• Centralised security (1) | • Award examples if mapped to a bullet point<br>• Do not award the same bullet more than once<br>• Do not award for install updates (in the stem) | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) (i) | Any **two** from:<br>• ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1)<br>• The connection does not get worse the further you are from the router / more reliable (1)<br>• Connection does not get obstructed by walls, ceilings, and furniture (1)<br>• More secure (1) | Accept reverse arguments for disadvantages of wireless. | 2 |

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) (ii) | Any **two** from:<br>• Installation/maintenance is more complex (1)<br>• Devices need to be physically connected (1)<br>• Less portable / limited by length of cable (1)<br>• Limited number of devices can be connected (1)<br>• Some digital devices cannot use a wired connection (1)<br>• Trip hazard (1) | Accept reverse arguments for advantages of wireless. | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c) | Award **two** marks for a linked description that addresses the individual mark points:<br><br>The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1)<br><br>The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|



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Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 3 of 5 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(d) | Award **two** marks for a linked explanation such as:<br><br>Running out of IPv4 addresses/the number of (internet) connected devices has grown (1)<br><br>IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e)(i) | Star | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) (ii) | Any **one** from:<br><br>• easy to connect/remove new nodes (1)<br>• failure of one node/link does not affect the rest of the network (1)<br>• easy to detect the failure of one node/link (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) (iii) | Any **one** from:<br><br>• if central switch/hub fails then the whole network fails (1)<br>• performance and number of nodes that can be added depends on capacity of central switch/hub (1)<br>• can require more cable than some of the other topologies (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 3 of 7 · Question 1 part b.i · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(f)(i) | Any **two** from:<br>• SMTP (1)<br>• IMAP (1)<br>• POP/POP3 (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(f)(ii) | Any **two** from:<br>• passes the (reassembled) packets to the application layer (1)<br>• check if all packets have arrived (1)<br>• determine whether the contents are correct (1)<br>• requests resending of lost or damaged packets (1)<br>• reassembles packets in correct order/ into an email (1) | Do not accept receives from network layer (stem) | 2 |



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# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 3 of 7 · Question 1 part b.i · 2 marks

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(a) | Any **two** from:<br>• (Share) access to the Internet/WWW/broadband connection (1)<br>• (Internal) communication (using email/instant messaging/calendars) (1)<br>• Share files/data (1)<br>• Share peripherals/printers / hardware (1)<br>• Increases the amount of storage (since network storage devices can be attached) (1)<br>• Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1)<br>• Centralised backup (1)<br>• Centralised security (1) | • Award examples if mapped to a bullet point<br>• Do not award the same bullet more than once<br>• Do not award for install updates (in the stem) | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) (i) | Any **two** from:<br>• ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1)<br>• The connection does not get worse the further you are from the router / more reliable (1)<br>• Connection does not get obstructed by walls, ceilings, and furniture (1)<br>• More secure (1) | Accept reverse arguments for disadvantages of wireless. | 2 |

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part b.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) (ii) | Any **two** from:<br>• Installation/maintenance is more complex (1)<br>• Devices need to be physically connected (1)<br>• Less portable / limited by length of cable (1)<br>• Limited number of devices can be connected (1)<br>• Some digital devices cannot use a wired connection (1)<br>• Trip hazard (1) | Accept reverse arguments for advantages of wireless. | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c) | Award **two** marks for a linked description that addresses the individual mark points:<br><br>The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1)<br><br>The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|



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Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 · extract 3 of 5 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(d) | Award **two** marks for a linked explanation such as:<br><br>Running out of IPv4 addresses/the number of (internet) connected devices has grown (1)<br><br>IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e)(i) | Star | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) (ii) | Any **one** from:<br><br>• easy to connect/remove new nodes (1)<br>• failure of one node/link does not affect the rest of the network (1)<br>• easy to detect the failure of one node/link (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) (iii) | Any **one** from:<br><br>• if central switch/hub fails then the whole network fails (1)<br>• performance and number of nodes that can be added depends on capacity of central switch/hub (1)<br>• can require more cable than some of the other topologies (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(f)(i) | Any **two** from:<br>• SMTP (1)<br>• IMAP (1)<br>• POP/POP3 (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(f)(ii) | Any **two** from:<br>• passes the (reassembled) packets to the application layer (1)<br>• check if all packets have arrived (1)<br>• determine whether the contents are correct (1)<br>• requests resending of lost or damaged packets (1)<br>• reassembles packets in correct order/ into an email (1) | Do not accept receives from network layer (stem) | 2 |



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Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 4 of 7 · Question 1 part b.ii · 2 marks

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a) | Award **one** mark for correct network type<br>Award **one** mark for correct usage model<br>One mark for each column | | 2 |

| Network type | Tick (□) | Usage model | Tick (□) |
|---|---|---|---|
| Local Area Network (LAN) | | Client-Server | |
| Wide Area Network (WAN) | | Peer-to-peer | □ |
| Personal Area Network (PAN) | □ | | |

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)</td><td>Award one mark for each correct layer name up to a maximum of two<br>Award one mark per layer for a correct function associated with the correct layer up to a maximum of two<br><br>
<table>
<tr><th>Layer</th><th>Function</th></tr>
<tr><td>Application</td><td>• Selects and uses the correct protocol to transmit data<br>• Interacts with the user</td></tr>
<tr><td>Transport</td><td>• Identifies the server port to use<br>• Identifies the client port to use<br>• Divides the data into packets<br>• Numbers the packets<br>• Adds the total number of packets<br>• Sets up communication between hosts / Establishes end to end communication<br>• Passes the packets to the network layer<br>• Checks the packets arrive at the destination<br>• Resends any packets that have not arrived</td></tr>
<tr><td>Network</td><td>• Adds the source/sender's IP address<br>• Adds the destination IP address<br>• Routes the packets</td></tr>
<tr><td>Data link</td><td>• Controls physical connections between pieces of hardware<br>• Adds MAC addresses to the packets<br>• Sends the packets on their way<br>• Adds headers and trailers</td></tr>
</table></td><td>Accept functions at the receiving end too</td><td>4</td></tr>
</table>

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(c) | **The only correct answer is A**<br><br>B *is not correct because* it is not a smartphone frequency band<br>C *is not correct because* it is not a smartphone frequency band<br>D *is not correct because* it is not a smartphone frequency band | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(d)(i) | Award **one** mark from:<br><br>• higher frequency has more waves per second/carries more data in the same time (1)<br>• higher frequency gives greater bandwidth (1)<br>• higher frequency is more stable/less prone to interference (1) | | 1 |
| 3(d)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1)<br>• gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1)<br>• gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1) | | 2 |

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(i) | Award **one** mark from:<br><br>• do not leave the smartphone on standby/turn off when not being used (1)<br>• use (portable) solar cells/other renewable generation method (1)<br>• unplug charger when not in use (1)<br>• reduce screen brightness (1)<br>• turn off WiFi/GPS/Bluetooth/location (when not in use) (1)<br>• close apps that are not being used (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(ii) | Award up to **two** marks for a linked explanation.<br><br>Award **one** mark for the action<br>• keep smartphone for longer (1)<br>• repair/recycle smartphone if not working (1)<br>• upgrade the operating system (1)<br><br>Award **one** mark for the impact<br>• less e-waste is generated (1)<br>• manufacturing could be reduced (1) | | 2 |
| | | **Total** | **13** |

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 5 of 7 · Question 3 part c · 1 mark

2021 NOV · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 6 of 7 · Question 6 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a)(i) | Award up to **two** marks for a linked explanation such as:<br>• Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1)<br>• No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1)<br>• Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1)<br>• Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1)<br>• Easy to find faults (1) all of the tokens will end up on one workstation (1)<br>• Cheap to set up (1) uses minimum cabling (1) | Not a **Comparison** without expansion that fits mark scheme | 2 |
| 6(a)(ii) | Award up to **two** marks for a linked explanation such as:<br>• It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1)<br>• Can handle high volumes of data traffic (1), because data can travel via multiple routes (1)<br>• Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc<br>• Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(b)(i) | Award **one** mark for:<br>• Personal area network / PAN/WPAN | | 1 |

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 6 of 7 · Question 6 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(b)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1)<br>• Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(c)(i) | Award **one** mark from:<br><br>• A record of activities/specified activity that have taken place on a computer system (1)<br>• Automatic record of what has happened and who did it (1) | | 1 |
| 6(c)(ii) | Award **one** mark from:<br><br>• To identify suspicious/malicious activity/changes (1)<br>• To increase accountability (1)<br>• To trace a problem back to its source/perpetrator (1)<br>• To find out if any users are using unauthorised applications (capable of putting the network at risk) (1) | | 1 |

Type MS · Wired or wireless? Media, pros and cons, and network speed units

Answer 6 of 7 · Question 6 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(d) | **Ethical hacking**<br>• Ethical hackers are white hat hackers<br>• Attempt to access the network as a hacker does<br>• Don't attempt to change or steal data<br>• Looking for weaknesses in the network<br>• Weakness pointed out<br>• Weaknesses fixed<br>• Could be employed by the business<br>• Could work for another specialist company<br>• Can include penetration testing<br><br>**Commercial analysis tools**<br>• Software used to find weaknesses<br>• Can be configured to check for a range of weaknesses<br>• Results/reports generated identifying faults<br>• Weaknesses fixed<br><br>**Review of network and user policies**<br>• Collection of rules and guidelines that govern the behaviours of network devices/users<br>• Need reviewing because may not comply with new laws and regulations<br>• Reviews should be scheduled | | 6 |
| | | **Total for question 6** | **15** |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1–2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured. |

## Type MS · Wired or wireless? Media, pros and cons, and network speed units

### Answer 6 of 7 · Question 6 part b.ii · 2 marks

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# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 7 of 7 · Question 4 part a · 1 mark

2023 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a) | **The only correct answer is C**<br><br>***A*** *is not correct because* ***mebibits*** *is a measure of storage capacity*<br><br>***B*** *is not correct because* ***mebibytes*** *is a measure of storage capacity*<br><br>***D*** *is not correct because transmission speeds are measured only in bits per second, not* ***bytes per second*** | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(b) | Award **one** mark for each correct cell: | | 2 |

| Type | Example | Number of bits |
|---|---|---|
| IPv4 | 192.169.0.3 | 32 (1) |
| IPv6 | 1050:a500:00c0:0440:0006:0300:700d:436f | 128 (1) |

# Type MS · Wired or wireless? Media, pros and cons, and network speed units

## Answer 7 of 7 · Question 4 part a · 1 mark

2023 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

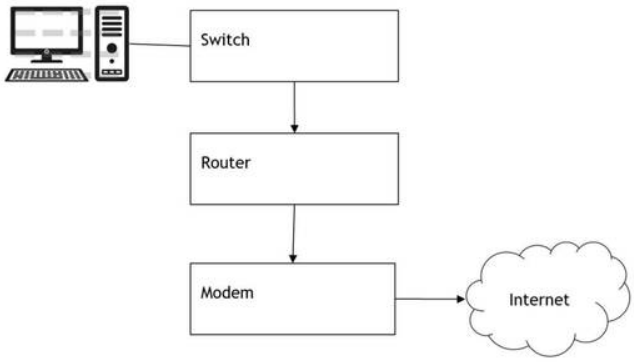
| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(i) | Award up to **two** marks for a linked description, such as:<br>• In a wired network packets travel over (copper/fibre-optic) cables (1), whereas in a wireless network packets travel over the air (1) | Allow radio signals, infra-red, microwave for air (must be more than just waves)<br><br>Allow twisted pair, CAT for copper.<br>Allow electrical/light signals (not just electricity) | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>1. A protocol (suite/group/family) (1)<br>2. Used on a wired network / wired connection (1)<br>3. Defines physical parts/type of cable(twisted pair, CAT6)/type of connector (1)<br>4. Defines how packets are checked for errors (1)<br>5. Defines the speed of transmission (1)<br>6. Operates at the link layer of the TCP/IP stack (1) | | 2 |

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## Answer 7 of 7 · Question 4 part a · 1 mark

2023 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(d) | Award **three** marks for all of switch, router and modem in the correct order as in diagram<br><br>Award **two** marks for all of switch, router and modem in any order<br><br>Award **one** mark for any of switch or router or modem in any box<br><br>  | | 3 |

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## Answer 7 of 7 · Question 4 part a · 1 mark

2023 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(e) | Award up to **two** marks for a linked description, such as:<br><br>Validation is used to check that the data/password entered by a user is fit for processing/meets specified requirements (1), whereas authentication is used to verify the identity of the user/prove that users are who they say they are (1) | For validation 'specified requirements' a good description of the check can be given e.g. presence check, length check, format check | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|